

CausalVerify v11 Execution Pipeline in Full Detail

CausalVerify v11 执行全流程详解

Bilingual Technical Walkthrough

中英双语技术指南

The complete pipeline has 6 stages. Below I explain step by step **what we do, how we do it, and what we produce**.

完整的 pipeline 分 6 个阶段，下面一步一步说明做什么、怎么做、产出什么。

1 Stage 0 —Data Preparation / 数据准备

Purpose: establish an evaluable input (question + true answer).

目的: 建立可评估的输入（问题 + 真实答案）。

1.1 Exp A: Real Papers / 真实论文

Script / 脚本: src/pipeline/expand_benchmark.py + src/pipeline/generate_paper_jsons.py

What it does / 做什么:

1. Call the OpenAlex API to query papers from top economics journals.
调用 OpenAlex API 查询顶级经济期刊论文
2. Use GPT-4o to generate `research_question` / `data_description` / `institutional_context` from the metadata.
GPT-4o 根据元数据生成三个关键字段
3. Human review and extension → 262 paper JSON files.
人工审核 + 扩展 → 262 篇 paper JSON

Input / 输入: 3-D allocation table of (economics domain × method × difficulty).

经济学领域 × 方法 × 难度 3D 分配表

Output / 输出:

```
experiments/exp_a/papers/paper_XX_did_easy.json (262 files)
Each contains:
- paper_id, authors, year, journal
- research_question shown to LLM
- data_description shown to LLM
- institutional_context shown to LLM
- ground_truth: hidden from LLM, used for scoring
```

```

method_family (DID/EVENT_STUDY/IV/RDD)
conclusion_direction (positive/negative)
main_effect
key_robustness

```

1.2 Exp B: Synthetic DGP Scenarios / 合成 DGP 场景

Script / 脚本: src/pipeline/generate_exp_b_scenarios.py + extend_exp_b_to_100.py

What it does / 做什么:

1. For 4 methods (DID/ES/IV/RDD), synthesize CSV data with NumPy so that the true β^* is completely known.
对 4 种方法，用 NumPy 合成 CSV 数据，真实 β^* 完全已知
2. Use GPT-4o to generate natural-language research questions (hiding the method name).
GPT-4o 基于合成参数生成自然语言 research question（隐藏方法名）
3. 30 hand-written + 70 programmatically extended = 100 scenarios.
30 个手工 + 70 个程序化扩展 = 100 个场景

Key design / 关键设计: seed=42 fixed, fully reproducible. / 固定 seed，完全可复现。

Output / 输出:

```

experiments/exp_b/scenarios/s01.json (100 scenario files)
- scenario_id, method_family, difficulty
- research_question, data_description
- data_file: "experiments/exp_b/data/sXX_data.csv"
- dgp_truth:
  type:      "did" / "event" / "iv" / "rdd"
  effect:    -0.3      <- exact true effect
  direction: "negative"

experiments/exp_b/data/s01_data.csv (100 CSV files)
1600 rows (200 unit x 8 period) x 6 columns

```

2 Stage 1 —LLM Generation / LLM 生成

Purpose: have each LLM produce R code + research plan for each task.

目的: 让每个 LLM 对每个任务生成 R 代码 + 研究方案。

2.1 Exp A Calls / Exp A 调用

Script / 脚本: src/pipeline/run_exp_a.py

What it does / 做什么 (for each paper $\times$ each model):

```
# System prompt
"You are a financial economist... Propose an identification strategy..."

# User prompt (fill in three fields / 填入三个字段)
"""
Research Question: {research_question}
Available Data: {data_description}
Institutional Context: {institutional_context}

Please provide:
1. Identification Strategy
2. Treatment and Control
3. Key Identifying Assumptions
4. Threats to Identification
5. R Code Outline
6. Expected Results
7. Validity Checklist
"""

-> Call LLM API -> save
```

Output / 输出:

```
experiments/exp_a/outputs/paper_XX_claude-opus-4-6.json (1,572 files)
Contains:
- paper_id, model, run_at
- llm_response:
  model, input_tokens, output_tokens
  content: complete markdown answer (with R code block)
  stop_reason
```

Scale / 规模: 262 papers $\times$ 6 models = **1,572 outputs**

Time / 时间: $\sim$ 6–12 hours (rate-limited) / 约 6–12 小时

Cost / 成本: $\sim$ \$80–150

2.2 Exp B Calls / Exp B 调用

Script / 脚本: src/pipeline/run_exp_b.py

Differences from Exp A / 与 Exp A 的区别:

1. Provide the LLM with the **first 5 CSV rows + the file path**.
给 LLM 提供 CSV 前 5 行预览 + 文件路径
2. The prompt requires **complete, runnable R code** (not an outline).
要求完整可运行的 R 代码
3. Compress the prompt to 3 sections. / 压缩到 3 节。

```
# User prompt template
"""
Research scenario: {title}
Research question: {research_question}
Data description: {data_description}
Data file path: experiments/exp_b/data/sXX_data.csv
Data preview (first 5 rows):
{pandas.to_string}
Column types: {dtypes}

Please implement the full analysis in 3 sections:
1. Identification Strategy
2. R Code (complete, self-contained)
3. Expected Findings
"""
```

Output / 输出:

```
experiments/exp_b/outputs/sXX_claude-opus-4-6.json (600 files)
Same structure as Exp A. Key difference: the R code in
'content' references a real CSV path that can later be
executed by L2b+.
结构同 Exp A。关键区别：R 代码的 CSV 路径真实存在，
可被后续 L2b+ 真正执行。
```

Scale / 规模: 100 scenarios $\times$ 6 models = **600 outputs**

Time / 时间: $\sim$ 3–5 hours / 约 3–5 小时

Cost / 成本: $\sim$ \$40–80

3 Stage 2 —External Verification (Pillar A) = L2b+

Purpose: verify correctness via code execution + coefficient comparison. **This is the paper's Channel A.**

目的: 用代码执行 + 系数对比验证正确性。这是论文的 Channel A。

Script / 脚本: src/pipeline/score_l2b_plus.py

3.1 Step 2.1 —L1 Check / L1 检查

```
l1 = bool(content.strip()) # Non-empty output? / 输出非空?
```

3.2 Step 2.2 —L2a: Extract R Code / 提取 R 代码

```
# Regex match markdown code block / 正则匹配 markdown 代码块
pattern = r"```(?:r|R)\n(.*)\n```"
r_code = extract(content, pattern)
l2a = r_code is not None
```

3.3 Step 2.3 —L2b: Actually Execute Code / 真正执行代码

```
# Write temp R script -> execute via Rscript -> 60s timeout
# 写入临时 R 脚本 -> Rscript 执行 -> 60s 超时
with open("/tmp/tmpXXX.R", "w") as f:
    f.write(f"options(warn = -1)\nsuppressMessages({{\n{r_code}\n}})")

result = subprocess.run(
    ["Rscript", "--vanilla", "/tmp/tmpXXX.R"],
    capture_output=True, text=True, timeout=60
)
l2b = (result.returncode == 0)
stdout = result.stdout
```

3.4 Step 2.4 —L2b+: Extract Coefficient + Compare / 提取系数 + 对比真值

```
# Three-tier extraction strategy / 三级提取策略
# Tier 1: labeled pattern "DID coef: -0.2987"
# Tier 2: line-position pattern (regression table)
# Tier 3: fallback scan (lines containing "treat"/"did"/"post")

estimated = extract_coefficient(stdout, method=scenario["method_family"])

# Compare against DGP truth / 与 DGP 真值对比
true_effect = scenario["dgp_truth"]["effect"] # -0.3
rel_error = abs(estimated - true_effect) / abs(true_effect)
l2b_plus = (rel_error < 0.5) # +/- 50% tolerance
```

Output / 输出:

```
experiments/exp_b/l2b_plus_scores.csv (600 rows)
Each row:
  scenario_id, method, model, model_slug
  true_effect, estimated, rel_error
  L1, L2a, L2b, L2b_plus
  err_excerpt (if failed)

experiments/exp_b/l2b_plus_summary.json
{ "Opus": { "L1": 100, "L2a": 99, "L2b": 94, "L2b_plus": 50 }, ... }
```

Typical results / 典型结果:

Model	L2a	L2b	L2b+
Opus	99	94	50
GPT-4o	100	78	25
Sonnet	100	51	21
o3	100	50	11
Kimi	100	45	5
Gemini	57	32	13

What Channel A tells the paper / Channel A 告诉论文:

“External verification is feasible; L2b+ can precisely measure which model outputs are numerically correct.”

“外部验证 feasible; L2b+ 可以精确测量哪些模型输出 numerically correct。”

4 Stage 3 —Internal Verification (Pillar B) = Calibration

Purpose: measure whether the model's confidence in its own output matches actual correctness.

This is the paper's Channel B — new in v11.

目的: 测量模型对自己输出的自信程度是否与实际对错匹配。这是论文的 Channel B — v11 新增。

4.1 3.1 Pilot (just completed) / Pilot (刚做完)

Script / 脚本: src/pipeline/run_calibration_pilot.py

What it does / 做什么:

1. From l2b_plus_scores.csv, sample 10 L2b+ failures + 5 L2b+ passes.
从 l2b_plus_scores.csv 抽 10 个 L2b+ 失败 + 5 个 L2b+ 通过
2. For each case, re-query Opus with a retrospective prompt:
对每个 case 重新调用 Opus, 使用回顾式 prompt:

```
Here's your earlier analysis: {original_output}
```

```
Rate your confidence (0-1) on three dimensions:
```

- method_confidence
- specification_confidence
- numerical_confidence

```
Output JSON only. Do NOT reveal true value.
```

Output / 输出:

```
experiments/exp_b/calibration_pilot_results.csv (15 rows)
  scenario_id, method, L2b_plus,
  method_confidence, specification_confidence, numerical_confidence
```

Key numbers from the pilot / Pilot 的关键数字:

- Mean numerical_conf on wrong cases = **0.50** / 算错样本均值 = 0.50
- Mean numerical_conf on correct cases = **0.61** / 算对样本均值 = 0.61
- **Gap = +0.11** (supports “confidently wrong” hypothesis)
Gap = +0.11 (支持 “confidently wrong” 假设)

4.2 3.2 Full Experiment (next) / 全实验 (接下来要做)

What it does / 做什么:

1. Run calibration elicitation for all 100 scenarios × 6 models.
对 100 场景 × 6 模型全部做 calibration elicitation
2. Compute Expected Calibration Error (ECE). / 计算 ECE

3. Compute over-confidence ratio. / 计算过度自信比率

```
# Divide confidence into 10 bins / 将 confidence 分 10 个 bins
# For each bin:
#   actual pass rate = proportion of L2b+=1 in that bin
#   avg confidence = mean confidence within bin
# ECE = weighted average |actual pass rate - confidence|

# Over-confidence ratio:
P(conf > 0.8 | L2b+ = 0) / P(conf > 0.8 | L2b+ = 1)
```

Expected output / 预期输出:

```
experiments/exp_b/calibration_scores.csv (600 rows)
experiments/exp_b/calibration_summary.json (6-model summary)
  Per model: ECE_method, ECE_specification, ECE_numerical
              P(conf>0.8 | L2b+=0), P(conf>0.8 | L2b+=1)
              over_confidence_ratio

paper/figures/calibration_reliability.pdf
  6 subplots, confidence bins x actual pass rate
  Diagonal = perfect calibration
  Curve deviating = miscalibration
```

What Channel B tells the paper / Channel B 告诉论文:

“Internal verification fails; self-reported confidence cannot be used as a triage signal.”

“内部验证 fails; self-reported confidence 作为 triage signal 不可靠。”

5 Stage 4 —Cross-Channel Analysis

Purpose: cross-compare results from Channel A and Channel B to derive the paper’s core finding — “asymmetry.”

目的: 交叉对比 Channel A 和 Channel B 的结果, 得出论文的核心 finding — “asymmetry”。

5.1 4.1 Head-to-head Ranking

Scripts / 脚本: head_to_head_ranking.py + bootstrap_head_to_head.py

What it does / 做什么:

1. Use L2b+ pass rate as “ground-truth ranking.” / 以 L2b+ 通过率作为 GT 排名
2. Compute Kendall τ between each metric’s ranking and the GT. / 计算 Kendall τ
3. Bootstrap 10,000 times to get 95% CI. / Bootstrap 10,000 次得 95% CI

Output / 输出:

```
experiments/exp_b/head_to_head_ranking.json
{
  "L2b vs GT": { "tau": +0.73, "95CI": [+0.33, +0.87] },
  "L4_S1 vs GT": { "tau": -0.07, "95CI": [-0.41, +0.36] },
  "Calibration ECE vs GT": { ... } <- NEW
}
```

5.2 4.2 Asymmetry Analysis (new in v11) / Asymmetry 分析 (v11 新增)

What it does / 做什么: for each model, construct a “channel profile”: / 对每个模型, 构造 “channel profile”:

Model	L2b+ pass rate	ECE
Opus	0.50	0.34
GPT-4o	0.25	0.42
...	...	...

Then analyze: / 然后分析:

1. **Correlation** between Channel A and Channel B. / Channel A vs Channel B 的相关性
2. Whether any model is **good on Channel A but poor on Channel B** (= the paper’s asymmetry claim).
是否有模型 Channel A 好但 Channel B 差 (= 论文 asymmetry claim)
3. Scatter / PCA visualization of the “profile.” / 散点图 / PCA 展示

Output / 输出:

```
paper/figures/channel_asymmetry.pdf
Scatter plot (x = L2b+, y = 1 - ECE = calibration quality)
Diagonal: "two channels agree"
Off-diagonal points: asymmetry (your core finding)
对角线表示 "两 channel 一致", 点偏离对角线 = asymmetry
```

6 Stage 5 —Figure Generation / 图表生成

Scripts / 脚本: paper/figures/make_fig*.py

Figures to produce (v11 checklist) / 要生成的图 (v11 清单):

Figure	Script	Content
Fig 1	make_two_dim_framework.py	Two-dimensional framework (done)
Fig 2	make_fig2_headline.py	L2b+ cascade + model ranking
Fig 3	make_fig3_clean.py	Method-level dotplot
Fig 4	make_fig4_cascade.py	Execution funnel
Fig 5	make_fig_error_cdf.py	Relative error CDF
Fig 6	make_calibration_reliability.py	Reliability diagram (NEW)
Fig 7	make_channel_asymmetry.py	Two-channel scatter (NEW)

Each script / 每个脚本:

- Reads from experiments/exp_b/*.csv or *.json / 读取数据
- Uses matplotlib + palette.py / 用 matplotlib + 统一配色
- Outputs paper/figures/*.pdf + *.png / 输出 PDF + PNG

7 Stage 6 —Paper Compilation / 论文编译

7.1 6.1 LaTeX Source Editing / LaTeX 源码编辑

File / 文件: paper/latex/neurips_v11.tex

Section changes in v11 / v11 的章节改动:

- §2 renamed to **Two-Dimensional Evaluation Framework** (+1.5 pages new)
§2 改为 “Two-Dimensional Evaluation Framework” (新增 1.5 页)
- §3 adds **§3.2 Pillar B: Self-Assessment Calibration** (+0.5 pages new)
§3 新增 “Pillar B: Self-Assessment Calibration”
- §4 adds **§4.3 Calibration Results + §4.4 Cross-Channel Asymmetry** (+1.5 pages new)
§4 新增 Calibration Results + Cross-Channel Asymmetry
- §6 adds **Practitioner Implications** (+1 page new) / §6 新增 “Practitioner Implications”

7.2 6.2 Compilation / 编译

Docker approach / Docker 方式:

```
docker run --rm -v $(pwd)/paper/latex:/work -w /work \
  texlive/texlive:latest \
  bash -c "pdflatex neurips_v11.tex && bibtex neurips_v11 && \
    pdflatex neurips_v11.tex && pdflatex neurips_v11.tex"
```

Output / 输出: `paper/latex/neurips_v11.pdf` (~15 pages main + appendix)

8 One-Page Pipeline Overview / 完整 Pipeline 一页总览

STAGE 0 - Data Preparation / 数据准备

Exp A: OpenAlex -> 262 paper_XX.json

Exp B: NumPy seed=42 -> 100 sXX.json + 100 CSVs + beta* known

|

v

STAGE 1 - LLM Generation / LLM 生成

run_exp_a.py: 262 x 6 = 1,572 outputs (~\$100, 12h)

run_exp_b.py: 100 x 6 = 600 outputs (~\$60, 5h)

|

v

STAGE 2 - Pillar A (External Verification) = L2b+

score_l2b_plus.py: L1 -> L2a -> L2b -> L2b+

-> l2b_plus_scores.csv + summary.json

Key result: L2b+ pass rate 5-50%, tau(L2b+, GT) = +0.73

|

v

STAGE 3 - Pillar B (Internal Verification) = Calibration [NEW]

Pilot: 15 samples -> gap = 0.11 ('confidently wrong' holds)

Full: 600 calibration queries

-> calibration_scores.csv + summary.json

Key result (expected): ECE > 0.3, overconfidence asymmetric

|

v

STAGE 4 - Cross-Channel Analysis

head_to_head_ranking.py + bootstrap_head_to_head.py

+ asymmetry analysis (A vs B correlation per model)

-> head_to_head_ranking.json + channel_asymmetry.pdf

|

v

STAGE 5 - Figure Generation / 图表生成

Figures 1-7 (PDF + PNG)

|

v

STAGE 6 - Paper Compilation / 论文编译

Docker + pdflatex -> neurips_v11.pdf (~15 pages)

9 Each Stage’s “Paper Argument” / 每个 Stage 对应的“论文论点”

Stage	Paper section / 章节	Argument carried / 承担的论证
0	§3.1 Methodology	Dataset + ground truth construction
1	§4.1 Setup	How 6 models are tested
2	§4.2 Pillar A	Channel A is feasible ($\tau = +0.73$)
3	§4.3 Pillar B	Channel B fails ($\text{ECE} > 0.3$)
4	§4.4 Cross-Channel	Core finding: asymmetry
5	Figures	Visual evidence
6	§6 Discussion	Practitioner implications

10 Where You Are Now / 你现在处在哪一步

[DONE] Stage 0 - Data preparation (v10 already done)
 [DONE] Stage 1 - LLM generation (v10 already done)
 [DONE] Stage 2 - Pillar A (v10 already done)
 [PARTIAL] Stage 3 - Pillar B: pilot done, full pending <- YOU ARE HERE
 [TODO] Stage 4 - Cross-Channel analysis
 [TODO] Stage 5 - New figures
 [TODO] Stage 6 - LaTeX editing + compilation

11 Three Choices Ahead / 接下来的三个选择

1. **Scale to the full calibration experiment** (Stage 3 complete, Day 2–4) — \$30–60, a few hours.
 扩到全 calibration 实验 (Stage 3 完成, Day 2–4) — \$30–60, 几小时
2. **First improve the prompt and script** (more rigorous contemporaneous elicitation) — half a day.
 先改 prompt 和脚本 (更严谨的 contemporaneous elicitation) — 半天
3. **First write §2 Framework LaTeX** (Stage 6 ramp-up, no new data) — 2 hours.
 先写 §2 Framework LaTeX (Stage 6 起步, 不依赖新数据) — 2 小时